rem: random energy model: T_c, frozen 1RSB phase

1 Model

family = rem, variables = ising, disorder = gaussian, constraint = None, prior = None.

random energy model: T_c, frozen 1RSB phase [1, 2]

Hamiltonian and conventions [1, 2]

$$2^N \text{ i.i.d. energies } E_i \sim \mathcal{N}(0, N_s N), \quad N_s = 1/2 \text{ (Derrida's } J = 1\text{; default)} \tag{1}$$

2 Method and route

Route: **replica** (replica method: replicated partition function Z^n, disorder average, order parameters q_{ab} (and conjugates), saddle point, RS / 1RSB ansatz, continuation n $\to$ 0). Rigor: **physics-level**. physics-level: replica $n \to 0$ continuation and RS/1RSB ansatz (Parisi), not proven; numbers carry quadrature error only

3 Derivation

No step-by-step symbolic derivation is available in this package for this model; the governing equations implemented by the code are shown below, each in sympy form (checked against the code by `tests/test_latex.py`) or cited. No intermediate step the code does not perform is shown.

3.1 Equations implemented by the family code

Conventions e = E/N, N_s = s (energy variance per spin); entropy, β_c and the free energy as in `families/rem.py`

Microcanonical entropy [1, 2] (sympy form; checked against the code by `tests/test_latex.py`)

$$s(e) = -\frac{e^2}{2s} + \log(2) \tag{2}$$

Critical inverse temperature ($s(e_0) = 0, \beta_c = s'(e_0)$) (sympy form; checked against the code by `tests/test_latex.py`)

$$\beta_c = \frac{\sqrt{2}\sqrt{\log(2)}}{\sqrt{s}} \tag{3}$$

Paramagnet, beta $\leq \beta_c$ (annealed = quenched) (sympy form; checked against the code by `tests/test_latex.py`)

$$-\beta f = \frac{\beta^2 s}{2} + \log(2) \tag{4}$$

1RSB functional (blocks of m replicas share a configuration; n $\to$ 0) (sympy form; checked against the code by `tests/test_latex.py`)

$$-\beta f = \frac{\beta^2 m^2 s + \log(4)}{2m} \tag{5}$$

Frozen phase, beta $> \beta_c$ ($m = \beta_c/\beta$) (sympy form; checked against the code by `tests/test_latex.py`)

$$f = -\sqrt{2}\sqrt{s}\sqrt{\log(2)} \tag{6}$$

4 Calculation at the requested parameters

The run's trace, in order (skipped steps say why).

4.1 Step 1: rem-critical-temperature

Source: [1, 2]. Trace: set errors, `step_note`, values; computed `N_scale`, β_c, T_c

4.2 Step 2: rem-saddle

Source: [1, 2, 3]. Trace: set `step_note`; skipped: no beta given

5 Solution

Values at the requested parameters (none), each with its finality: only stable and exact-by-literature values are final.

key	value	finality (ansatz)	final
beta_c	1.6651092223154	exact-by-literature (1rsb) [4, 2, 5]	yes
T_c	0.600561204393225	exact-by-literature (1rsb) [4, 2, 5]	yes

Other outputs (parameters, labels, diagnostics): `N_scale` = 0.5.

6 Stability

No stability quantity (replicon / AT) is computed by this run.

exact-by-literature: β_c, T_c.

7 Reference check

Banked values (reference/*.json) against this pipeline's output; numbers are shown to 16 significant digits, the comparison uses them in full; slow and statistical rows run only with –full (`render_latex(slow=True)`).

key	computed	banked	tolerance	verdict	source
tc_derrida	0.6005612043932249	0.6005612043932249	1e-20	agrees	[4, 6, 7]
ground_energy_derrida	-0.8325546111576978	-0.8325546111576978	1e-20	agrees	[4, 8]
m_times_beta_over_beta_c	1.0	1	1e-20	agrees	[9, 10]

8 Labels

Rigor: **physics-level**. Route: **replica**.

physics-level: replica $n \to 0$ continuation and RS/1RSB ansatz (Parisi), not proven; numbers carry quadrature error only

References

[1] Derrida, PRL 45, 79 (1980)

[2] Derrida, Phys. Rev. B 24, 2613 (1981) (exact solution: microcanonical entropy s(e))

[3] 1RSB form: Mezard, Parisi & Virasoro (1987)

[4] Derrida, The random energy model, Physics Reports 67, 29-35 (1980), Sec. 4: 'T_c = J/2 sqrt(log 2)' with P(E) = (pi N J^2)^(-1/2) exp(-E^2/(N J^2)), Eq. (2)

[5] Derrida & Mottishaw, arXiv:1410.1432 (m = T/T_c)

[6] read from the copy at thp.uni-koeln.de/krug/teaching-Dateien/WS2011/Derrida1980.pdf. Same statement in Derrida & Mottishaw, J. Stat. Mech. (2015) P01021, doi:10.1088/1742-5468/2015/01/P01021, arXiv:1410.1432, Eq. (2). Full paper Derrida, Phys. Rev. B 24, 2613 (1981), doi:10.1103/PhysRevB.24.2613 (not fetched). The paper states the closed form

[7] value = its decimal at J = 1 (replicant N_scale = J^2/2 = 1/2)

[8] also Derrida & Mottishaw, arXiv:1410.1432, Eq. (3). Closed form, decimal at J = 1. In the frozen phase f = E_ground / N (entropy 0)

[9] Derrida & Mottishaw, J. Stat. Mech. (2015) P01021, arXiv:1410.1432, Eqs. (4)-(6): Parisi q(x) = Theta(x - 1 + <P2>) with <P2> = 1 - mu, mu = T/T_c below T_c, so the 1RSB breakpoint is m = T/T_c = beta_c/beta

[10] same in Derrida & Mottishaw, J. Phys. A 55 (2022), doi:10.1088/1751-8121/ac71e9, arXiv:2202.12584, Eq. (5) (<Y_2> = 1 - beta_c/beta). Meaning: m * beta / beta_c = 1 for beta >= beta_c